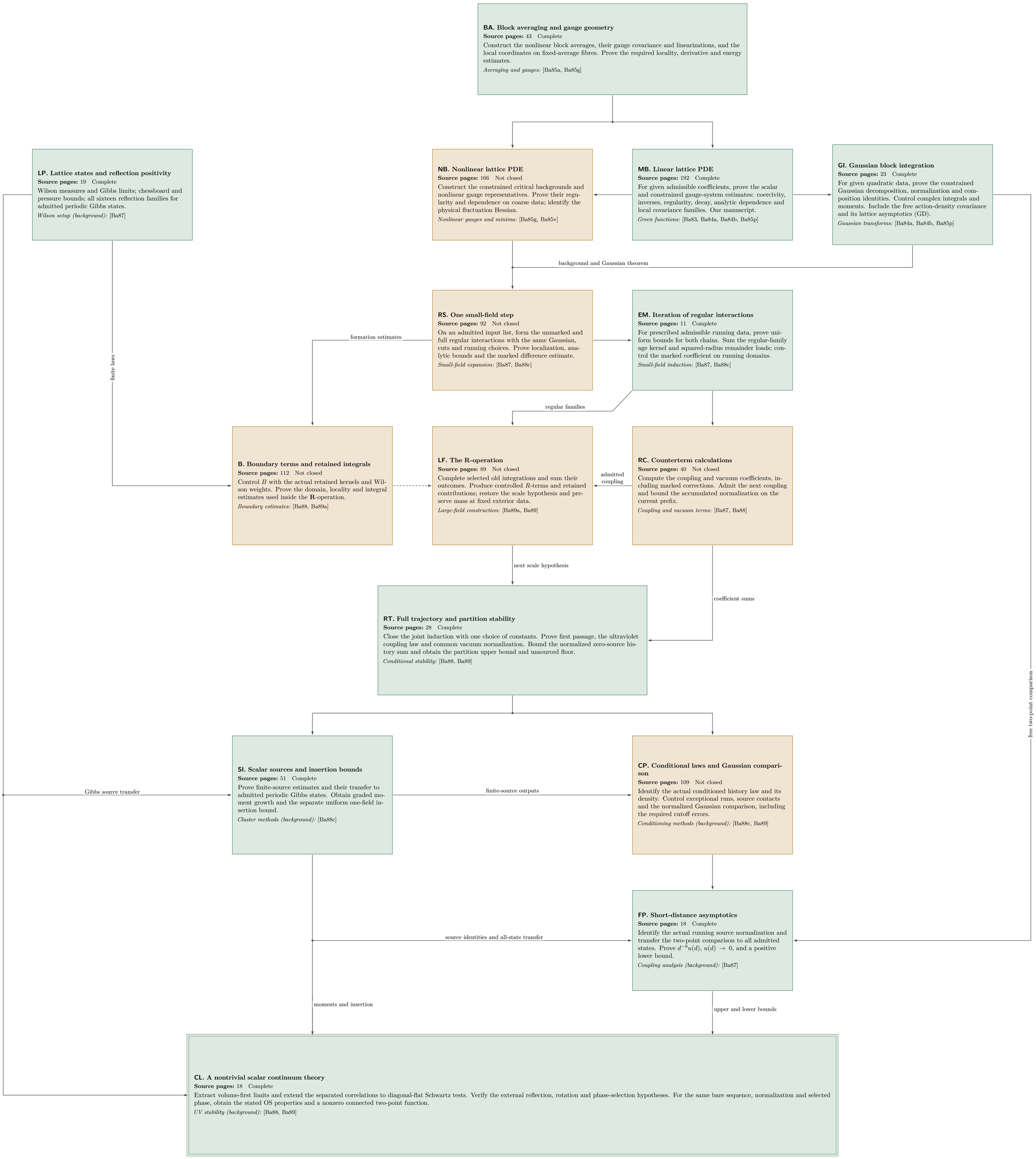
YM4 assembly outline

A scalar action-density theory for four-dimensional SU(2)

`YM4_YM4O` — Volume first, then a subsequential continuum limit.

Each heading links to its statement; each page count links to its source ranges and completion status. The literature labels link to precise passages; citations link to the bibliography. The 1,394-page source contributes 1,011 pages to these boxes and 383 pages to the other categories in §14. Every physical page is counted once.



Complete Not closed Not established / not assessed

Green means the box's argument is complete in the 1,394-page source, assuming its stated inputs, even if those inputs remain unproved. Amber means part of the box's own argument remains unfinished.

Solid arrows show principal inputs; the dashed arrow records estimates used inside a proof. Incoming inputs are required together. The double border marks the target. RS is one step; EM controls its iteration on given admissible data. RT constructs the full trajectory, including inherited remainders. General reflection, rotation and phase criteria are external inputs (§11).

YM4 ASSEMBLY OUTLINE

Abstract. We organize the volume-first construction of a scalar action-density theory for four-dimensional Wilson SU(2) gauge theory. Block averaging precedes one manuscript for linear lattice PDE and one for nonlinear lattice PDE. Gaussian integration includes the free observable calculation. One-step and iterated regular formation, boundary estimates, the large-field operation, counterterms and the full induction have separate tasks. Sources and two-point estimates lead to continuum assembly. The linked statements specify the scope of each task. Completion status is assessed in the 1,394-page source manuscript, assuming each box's stated inputs. The linked source-page counts retain all existing material, including unfinished tasks.

Contents

1. The target 2
2. Lattice states and reflection positivity 3
3. Block averaging and gauge geometry 4
4. Linear lattice PDE 5
5. Gaussian block integration 6
6. Nonlinear lattice PDE 7
7. Ordinary integration and the R operation 8
8. Renormalization constants and trajectory 11
9. Scalar sources and insertion bounds 13
10. The conditioned measure and two-point law 14
11. External inputs 16
12. The scalar continuum theory 17
13. Dependencies and outstanding estimates 18
14. Actual source-page counts 20
References 24

1. The target

Theorem 0 (Target, conditional on the estimates below)**.** For sufficiently small reference coupling, a volume-first subsequential continuum limit of the renormalized scalar action density has a nontrivial phase whose correlations satisfy the Osterwalder–Schrader properties on Schwartz tests flat on coincidence diagonals. ■□

The precise scope is given in CL in §12. At every fixed spacing we first choose an infinite-volume Gibbs state. We then let the spacing tend to zero along the first-passage sequence constructed in RT. The observable is the scalar action density associated with $\mathrm{tr}(F^2)$.

BA supplies averaging geometry to MB's linear theory and NB's nonlinear background construction. GI combines Gaussian block integration with the free action-density calculation GD; RS applies it to NB's physical Hessian.

RS proves one small-field step for both regular chains; EM controls their iteration on prescribed admissible running data. RT constructs the actual trajectory, including B, LF and RC, and proves partition stability. SI, CP and FP supply the observable estimates used in CL.

LP. Lattice states and reflection positivity

Complete; 19 source pages.

Inputs. The Wilson SU(2) model at fixed spacing $a > 0$ and coupling parameter $\beta > 0$, on compatible periodic tori and their admitted class of periodic infinite-volume Gibbs states.

Required result. Construct the normalized positive measures

$$Z^{-1} \exp\left(\frac{\beta}{2} \sum_p \operatorname{Re} \operatorname{tr} U_p\right) \prod_b \mathrm{d}U_b,$$

where U_b is a link variable, U_p is its oriented plaquette product and $\mathrm{d}U_b$ is Haar probability measure. Establish coordinate site/bond reflection positivity and subsequential infinite-volume Gibbs limits. The lattice symmetries and coordinate reflection forms pass to these torus-selected limits.

On tori adapted to a coordinate or diagonal mirror, prove the crossing-plaquette positivity, reflected-fold and chessboard inequalities, and pressure comparisons for repeated real local tests. Include the Gibbs/Jensen lower tangent. Boundary errors vanish at fixed a, β; the volume limit precedes the local perturbation limit.

Prove lattice translation and cubic symmetry and reflection positivity for every state in the admitted periodic Gibbs class, on bounded Borel half-space algebras. The mirror families are

$$x_\mu = c, \quad c \in \tfrac{a}{2}\mathbb{Z}; \qquad x_\mu \pm x_\nu = c, \quad c \in a\mathbb{Z}, \quad \mu < \nu.$$

There are four coordinate and twelve diagonal orientations. This extension requires pressure comparison for every real local test, including mirror-straddling tests, the common-period and link-symmetry argument, arbitrary-stack layer removal, and separation in the compact convex comparison class. Weak closure of torus laws alone does not establish it.

Source locations. `YM4_YMSO`: G1; the finite Wilson model, adapted crossing kernels, chessboards, periodic link-symmetry and all-mirror chapters; the detailed YM4 outline, §6.3 and the Gibbs-reflection task in §9.4.

Bałaban references. The compact-link Wilson action on the torus is specified in [Ba87, pp. 251–252, (0.2)]. This is background for the lattice model; the all-state reflection argument is the one identified above.

BA. Block averaging and gauge geometry

Complete; 43 source pages.

Inputs. The fine and coarse lattices, the specified block geometry, SU(2) link variables and path conventions. For local analytic statements, impose the source's near-identity or small-curvature conditions.

Required result. Construct the actual group-valued block average and prove its gauge covariance, locality and stated symmetry properties. Establish the derivative and remainder estimates needed when it is iterated, and the curvature and Wilson-energy bounds under their stated hypotheses.

Construct the local transport and axial gauges, the straightened averaging coordinates and the associated site and link rows. Prove the relevant composition and gauge identities. Define the local pivot chart on a fixed-average fibre, with its radius, Jacobian and analytic continuation. These finite local constructions precede the PDE estimates.

The linear rows and their estimates are supplied to MB; the nonlinear map and fibre geometry are supplied to NB and integration. Solving a nonlinear Landau-gauge equation or constructing a critical background belongs to NB. Bounds for inverses of the resulting linear operators belong to MB. The physical constrained Hessian uses these geometric data, but its identification is made in NB.

Source locations. `YM4_YMSO`: The averaging and geometric parts of §3.1, printed pp. 42–76; the energy estimate in §3.2, pp. 76–80; the pivot chart and local measure formula in §3.3, pp. 80–85. These sections also contain PDE and integration applications, which are assigned to their respective tasks.

Bałaban references. Iterated curvature bounds, straightened averages and their analytic estimates are developed in [Ba85a, §§B–F, pp. 23–51], especially Propositions 1–7. The axial and Landau conventions are given in [Ba85g, §B, pp. 78–82]. The chosen centered average and pivot chart require the separate calculations in `YM4_YMSO`.

MB. Linear lattice PDE

Complete; 192 source pages.

Inputs. BA's maps and linearized rows, the separated lattice geometry, physical pairings and boundary conventions, and a given field/current pair in the specified real or complex coefficient class. Each class is defined by its inequalities, independently of NB's construction. For the physical fluctuation estimates, supply the stated constrained Hessian data, including the constraint-curvature and multiplier terms.

Required result. Prove the required linear estimates in one manuscript, with the following internal divisions.

(i) Scalar elliptic estimates: heat and Green kernels, finite-p and Hölder regularity, scalar block gaps and covariant local estimates.

(ii) Gauge constraints and quadratic forms: row lifts, gauge projections, Gram matrices, flat and regular-background coercivity, and the constrained field–multiplier systems. Prove the physical gap after removal of the artificial row mass under the specified Hessian hypotheses.

(iii) Green functions and reconstruction: the Green operators for the scalar elliptic equations and constrained field–multiplier systems, coarse Schur operators, current corrections, decay, strong regularity and complex continuation.

(iv) Local operator families: connected inverse and reconstruction families, changes of determining region, conditional covariance matrices and the required matrix functions on their admitted complex domains.

Retain each operator's own derivative range, boundary convention and coefficient hypotheses. Uniformity is in mesh, depth and admitted volume. Bounds for a summed kernel do not replace bounds for its local families. NB constructs the actual coefficients and identifies the physical Hessian; the linear estimates are conditional on the stated input data.

Source locations. `YM4_YMSO`: The scalar, constrained-inverse and local-family arguments in §3 and Appendix A.1; the linear parts of §5.11–5.13. The nonlinear response on pp. 201–205 and the application-domain checks on pp. 240–241 are excluded.

Bałaban references. The scalar covariant Green-function estimates are stated in [Ba83, pp. 573–574]. Constrained gauge propagators and their mixed-scale extensions are treated in [Ba84a, §§D–F, pp. 26–40] and [Ba84b, §§A–C, pp. 223–249]. For regular external backgrounds, use [Ba85p, Theorems 3.1–3.4, pp. 397–400; §§C–E, pp. 408–432]: these supply analytic inverse bounds, local walk expansions, constrained corrections and fluctuation covariances. Each result retains its field, geometry and output hypotheses. The covariant-Laplacian bound in (3.42) is explicitly excluded in [Ba85p, Theorems 3.12–3.13, pp. 423 and 426].

GI. Gaussian block integration

Complete; 23 source pages.

Inputs. A specified quadratic action, linear averaging constraints, gauge and measure conventions, and the coercivity, inverse and local matrix estimates from MB. For complex data, use the stated sector and parameter domains of those estimates.

Required result. Identify the constrained minimizer, the centered fluctuation covariance and the effective coarse quadratic action. Prove the exact Gaussian block-integration formula, retaining its measure and gauge normalizations. Establish compatibility with successive block integrations.

Continue the determinant branch from the positive real forms. Prove the normalized complex integral, moment and parameter-derivative bounds on the admitted domains, with the stated cost per localization cube. Import the operator estimates from MB. This conditional theorem takes the quadratic data as given. NB identifies them for the physical background; the nonquadratic action, Jacobian and cutoff corrections are treated in RS.

Free action-density covariance (GD). For the flat Gaussian curvature, compute the Wick contraction and its positive coefficient C_G. Establish the separated lattice comparison with

$$\mathcal{K}(f,h) = \iint_{\mathbb{R}^4\times\mathbb{R}^4} f(x)h(y)|x-y|^{-8}\,\mathrm{d}x\,\mathrm{d}y$$

on the test class required by FP. The interacting comparison and physical source normalization remain in CP and FP.

Source locations. `YM4_YMSO`: The free decomposition in Proposition 3.21, pp. 63–64; the Gaussian composition identity in §5.3.6, pp. 866–867; normalized complex integrals in §5.12.5, pp. 913–914; the free parts of F1, pp. 1015–1016, and Appendices A.3–A.4, pp. 1267–1277. The nonlinear Hessian identification in the composition argument belongs to NB.

Bałaban references. The Gaussian block transform and its composition are given in [Ba84a, §§A–B, pp. 17–20]; its constrained shift and normalization are computed in §D, pp. 26–29, especially (1.64)–(1.65). Mixed-scale restrictions appear in [Ba84b, §A, pp. 223–229], and the background-field fluctuation covariance in [Ba85p, §E, pp. 427–432, Theorem 3.15]. The complex-sector and free composite-density calculations use the additional arguments in `YM4_YMSO`.

NB. Nonlinear lattice PDE

Not closed; 166 source pages.

Inputs. BA's nonlinear averages, local gauge coordinates and fibre charts; MB's conditional inverse and regularity estimates; regular coarse data and the boundary conditions of the constrained problem.

Required result. Construct the constrained critical backgrounds and the nonlinear gauge representatives required by the construction. Prove their regularity, analytic dependence on the coarse data, local comparison and determining-region estimates, on the stated real and complex domains. Treat the nonlinear field–multiplier response equations here, using the linear estimates from MB.

Verify that the constructed pairs lie in the coefficient classes used by the relevant MB theorems. Differentiate the actual constrained action in BA's coordinates to identify its physical fluctuation Hessian, retaining the second derivative of the constraint and the multiplier terms. MB then supplies the corresponding gap and covariance estimates. This is application of a conditional linear theorem, not a reverse prerequisite in the construction of that theorem.

State precisely which nonlinear response domains and substitutions the theorem controls. A later history, source perturbation or retained integral must verify those hypotheses at its own application. The complete history-dependent domain checks remain with RS, B and LF; they are not consequences of critical-background existence alone.

Source locations. `YM4_YMSO`: The nonlinear gauge and critical-background parts of §3, including the coupled response on pp. 201–205; the physical Hessian identification in §5.11 and §5.13.

Bałaban references. The nonlinear gauge transformation is constructed in [Ba85g, Theorems 2, 4 and 8, pp. 83, 88 and 101]. The regular constrained minimal orbit and its uniqueness within the admitted class are the subject of [Ba85v, Theorem 1, p. 279]; analytic dependence and response expansions are discussed in its §F, pp. 300–309. The kernel calculation leading to (189) is not carried out there (p. 308), so that passage alone cannot supply all the response bounds required here.

7. Ordinary integration and the R operation

Write H_k for the hypothesis on the current scale-k measure presentation: its local E, R, B functions, retained positive kernels, their component factorization and ordered composition, domains, locality and quantitative bounds. It describes present objects without prescribing how they were constructed. Older remainder functions are part of this input.

RS. One small-field step

Not closed; 92 source pages.

Inputs. BA, MB, NB and GI; common reference and running choices, actual action weights, tests, radii and localization. Supply an admitted list of earlier regular families with their action and vacuum subtractions, and eligible vacuum-subtracted inherited remainders, together with their companions and analytic bounds.

Required result. Form the unmarked output and the full regular output by one small-field integration. The unmarked input keeps only the unmarked regular families. Both calculations use the same Gaussian, cuts, reference data and compatible domains. Their difference defines the marked output.

Use NB's physical Hessian identification and GI's Gaussian formula on the actual averaging fibre. Retain the pivot, coarea, Haar and gauge factors. Treat the nonquadratic action and Jacobian terms with the actual frozen cutoffs. Separate the Gaussian normalization and the action/reference-ratio correction from the small scalar correction. Prove formation, localization and analytic bounds, including the enlarged domains and strengthened decay used by the next operation. The stronger $e^{-(1+4\beta_s)\kappa d(X)}$ estimate applies to the localized scalar correction; Gaussian families have their separate bounds.

Compare the two frozen integrals to bound the marked difference by the marked regular inputs and eligible old remainders. Retain the summable regular-family age kernel and the squared ratio of evaluation radius to creation radius for each remainder. Prove the actual common-cut and domain compatibility required by this comparison. The output is formed before choices that use the next reference entry. RC subsequently checks the computed next coupling. The same-scale **R**-operation supplies no input to this step. Younger remainders retain their separate R/B outputs. With B's boundary and retained-kernel estimates, this gives the regular part of the intermediate ordinary pushforward.

Source locations. `YM4_YMSO`: B1–B4; finite formation in §4.5 and §4.15; the one-step functional and input-list load in §5.2.1, printed p. 832; the common frozen-integral comparison in §5.2.3, p. 841.

Bałaban references. The fluctuation integral and its analytic expansion are treated in [Ba87, §§2–3, pp. 265–281]. Localization and the exponentiated cluster expansion are completed in [Ba88c, §§1–2, pp. 2–22]; the stronger pre-R margins are recorded in [Ba88, p. 262]. The marked comparison and the current joint-domain and cutoff checks remain the tasks specified above.

EM. Iteration of regular interactions

Complete; 11 source pages.

Inputs. RS's one-step theorem; prescribed admissible reference and running data on a finite prefix; inherited remainders satisfying their creation bounds, age gates, companions and holomorphy hypotheses. Supply the flow and radius comparisons needed for the marked remainder sum.

Required result. Iterate the unmarked and full regular constructions and set

$$E_j^{\mathrm{m}} = E_j - E_j^{\mathrm{u}}.$$

Prove bounds uniform in the number of formed steps, on each chain's stated domains. The unmarked recursion starts from the empty list and uses absolute radii. The full recursion uses the admitted aged remainders on running radii. Restriction between these domains must restrict the same actual functions, cuts and logarithm branches.

Sum the regular-family age kernel and control the inherited remainder loads with their squared-radius weights and the stated flow bounds. Close the marked weighted-supremum recurrence using RS's comparison estimate. Prove the running-radius Cauchy bound needed to extract the marked coupling coefficient, preserving the domain and companion compatibility along the prefix.

The theorem is conditional on the supplied running data and inherited remainder bounds. With those choices fixed, the unmarked chain has no R-inputs. EM neither constructs a coupling trajectory nor closes the B/R histories. RT verifies its hypotheses along the actual joint construction. The required age-kernel estimate must already be proved in RS.

Source locations. `YM4_YMSO`: The two chains and marked recurrence in §5.2.1, printed pp. 833–834; the common frozen-integral comparison and supremum induction in §5.2.3, p. 841. The first-source recurrence and all-state source bounds remain in SI.

Bałaban references. The conditional small-field induction is stated in [Ba87, Theorems 1 and 3, pp. 259 and 264], with hypotheses in §1, pp. 260–264. Its localized effective-action bounds are completed in [Ba88c, §2, pp. 11–22]. These results assume sufficiently small running couplings; the full/marked age-weighted recurrence is the additional argument in `YM4_YMSO`.

B. Boundary terms and retained integrals

Not closed; 112 source pages.

Inputs. The current presentation, BA, MB, NB, RS and EM, and the boundary functions and retained kernels supplied by the ordinary integration. For partial large-field stages, take their specified conditional kernels, restrictions, exterior data and carrying Wilson weights.

Required result. Prove the domain and locality estimates for boundary functions B, including their extensions in retained variables. Inherited functions keep their creation domains; every later evaluation must lie in the original domain. Establish component factorization and ordered composition for the retained integrations.

Bound the designated local subintegrals, with their actual tests and weights, and prove the factorwise oscillation and complex-stage estimates used inside LF. Keep a separate mass estimate only where the source supplies one. A partial integral may require its accompanying Wilson weight for control; a bound on an unweighted integral is not presumed. These are estimates for boundary contributions and conditional integrals, not another transformation of the measure. LF proves that its successive stages and new boundary functions satisfy the required bounds.

Source locations. `YM4_YMSO`: C1–C4; retained strips, collar reads, creation-domain bounds, ordered partial kernels, complex stage integrals and zone reserves.

Bałaban references. Localized analytic boundary families and their decay bounds are specified in [Ba88, §2, (2.40)–(2.42), pp. 261–262]. The partial integrations creating new boundary terms are developed in [Ba89a, §1, pp. 178–191, especially (1.61)]. The stored-coordinate domains and uniform retained-integral bounds of the present construction still require the checks stated above.

LF. The R-operation

Not closed; 89 source pages.

Inputs. The intermediate ordinary presentation, including the regular families from RS with the uniform bounds from EM, and their strengthened domains and decay, the estimates in B, and RC's admission of the next coupling.

Required result. Construct the operation on this presentation: perform the prescribed tests and partial integrations, integrate selected old interior fields at fixed exterior data, and insert the required new fields with normalized reference laws. Sum completed contributions into local R-terms; retain the appropriate kernels and B-terms on the other outcomes.

Prove the stage domain inclusions, geometric margins, locality, rarity and history-sum bounds, and their finite-source versions. The induction over preparation and completion stages belongs to this proof. It must produce the next presentation satisfying H_{k+1} with the original constants, using the stronger ordinary-step margins. Inherited functions retain their domains; new R and B functions and the remaining kernels satisfy their specified estimates.

Writing the output as $\mu_{k+1} = \mathbf{R}(T\mu_k)$, prove equality of total mass, also at each fixed exterior datum. The operation can change the current measure. Preservation of observable correlations requires their prescribed transport and source identities; equality of mass alone does not supply it. This is a conditional single-scale result. RT constructs a sequence on which it applies.

Source locations. `YM4_YMSO`: C1–C4; actual history closure; RF19–RF25; the positive assembly of completed and retained outcomes.

Bałaban references. The basic R-operation and its normalization identity are constructed in [Ba89a, (0.4), p. 176; §1, pp. 177–201]. Localization, exponentiation and the final R/B bounds are completed in [Ba89, §1, pp. 356–392], under the small-running-coupling hypothesis of its Theorem 1. Application to the current histories requires their domain and envelope estimates.

RC. Counterterm calculations

Not closed; 40 source pages.

Inputs. BA, MB, NB, the regular formation from RS and EM, and the remainder and history bounds already available for the given admissible prefix. Fix the extraction and normalization conventions. No next-scale closure or completed trajectory is assumed.

Required result. **Coupling.** Compute the Gaussian coefficient b_j^0, including gauge, constraint and background terms, and prove

$$\left|\sum_{j=1}^{n} b_j^0 - nc_0\right| \le C, \qquad c_0 = \frac{11N^2}{12\pi^2}\log L > 0, \quad N = 2. \tag{1}$$

Here $L > 1$ is the blocking factor. Bound the interaction and history corrections to b_j^0 on the same admissible prefixes. In the conventional Wilson normalization $\beta = 2N/g_{\mathrm{std}}^2$, the coefficient for g_{std}^{-2} is $c_0/(2N)$.

Vacuum energy. For the accumulated logarithmic normalization V_k, compute

$$V_0 = 0, \qquad V_{k+1} = V_k + e_k.$$

The increment e_k includes the Gaussian determinants, Haar and gauge-fixing constants, and the values of the regular and remainder terms at the trivial field. Count each constant once. Prove the Gaussian constant bounds and accumulated-vacuum estimate (RF61)–(RF62), together with the compatibility of the vacuum densities on the admitted tori.

The linearly extracted coefficients of the two chains add to the full coupling coefficient. The marked norm estimate from EM must be converted to its physical coefficient with the same extraction convention.

Both calculations and their conditional accumulated bounds belong here. RT constructs an actual sequence to which they apply; RT uses the resulting common subtraction for partition stability. The next coupling must be admitted before LF applies the **R**-operation. There is no fermion mass parameter in this pure gauge model.

Source locations. `YM4_YMSO`: B3; RF10–RF12c and RF19–RF20; RF61–RF62; CP1; the common vacuum-density comparison.

Bałaban references. Ward identities and the polarization reduction defining the marginal coupling coefficient are developed in [Ba87, §§4–5, pp. 281–298; (1.20)–(1.22), p. 264]. Vacuum subtraction and its accumulated bound are given in [Ba88, pp. 254, 262 and 264]. These are references for the renormalization mechanism: the perturbative tuning theorem is announced with proof deferred in [Ba87, Theorem 2, p. 259], and [Ba89, p. 355] explicitly retains the running-coupling hypothesis.

RT. Full trajectory and partition stability

Complete; 28 source pages.

Inputs. RS, EM, B, LF and RC; an initial Wilson presentation satisfying H_0; and the actual zero-source history and normalization bounds.

Required result. Construct the joint coupling and vacuum-energy trajectory with one order of choice of constants. Form the unmarked and marked regular interactions, admit the computed coupling using RC, then restore the scale hypothesis by LF. Prove first passage into the terminal window, arbitrarily fine admissible cutoffs and the logarithmic ultraviolet law. Individual full coupling increments need not be positive.

With the common vacuum subtraction, bound the normalized zero-source history sum and all remaining weighted integrals. Establish the reference-volume partition upper bound and the unsourced floor, uniformly in cutoff at the fixed small reference coupling. This uses the actual component majorants, label counting and all-small-field lower-bound event. Equality of mass through the operations alone does not prove stability. Complex-source estimates and Gibbs-state transfer are proved in SI.

Source locations. `YM4_YMSO`: E1; RF22–RF27a; CP1; C4; the unsourced specialization of §5.21 and the floor in §4.20.

Bałaban references. The effective-density and ultraviolet-stability statements are [Ba88, Theorem 2 and Corollary 3, pp. 263–264] and [Ba89, Theorem 1 and (0.1), pp. 355–356]. They retain assumptions on the running couplings. They provide the conditional partition framework; the actual common trajectory and coefficient matching are the additional requirements of RT.

Bałaban's effective-density and stability arguments supply the renormalization framework. His large-field paper leaves physical observable expectations for further analysis [Ba89, p. 356]. The scalar source and two-point tasks below require the additional arguments in `YM4_YMSO`; they are not conclusions of that stability theorem.

SI. Scalar sources and insertion bounds

Complete; 51 source pages.

Inputs. RT's admitted run and partition bounds; LP's finite reflection, chessboard and Gibbs-state results; the conditional local-source formation and retained-integral bounds from RS, EM, B and LF.

Required result. Construct the actual action-density source presentation with the reference data frozen as prescribed. Prove the ordinary complex-source upper bound, local holomorphic source energies and common normalizers, including source locality, entropy margins and the precise source transport identities. Each local member retains its own extracted action coefficient. The unsourced partition bound from RT does not supply these estimates.

Use LP and the unsourced floor to transfer the finite-torus source bounds to every admitted periodic Gibbs state by taking the volume limit at fixed cutoff. Retain one state-independent cell normalizer and control differences of state means.

Deduce graded separated-moment growth and, by a separate reflected seminorm argument, the uniform one-field insertion bound. Its constant must be independent of the number and degrees of the surrounding fields. Factorial moment growth alone is not the insertion estimate.

Source locations. `YM4_YMSO`: D1–D2; source formation and transport; §5.20.1, §5.22–5.23 and §6.9.

Bałaban references. The relevant methodological background is the localization and connected cluster expansion in [Ba88c, §§1–2, pp. 2–22].

CP. Conditional laws and Gaussian comparison

Not closed; 109 source pages.

Inputs. The real positive history representation and domain estimates from the one-step construction, RT's reference run and normalization, SI's finite-source outputs, and the actual background estimates from MB and NB, GI's Gaussian normalization, and its application in RS.

Required result. **The conditioned law.** Identify the actual resummed positive zero-source measure on a reference run window, its exterior sigma-field and observable pullbacks. Construct a common admitted chart with the full constrained density: coarea and Haar factors, gauge factors, complete Hessian, retained variables, boundary terms, old activities and exterior force. A Hessian bound alone does not identify this law.

Sum positive histories of the same physical event before normalizing. Remove exceptional runs in the required averaged moment norms, with errors small relative to the g^4 signal. Control exterior means and their covariances, and retain the mixed source-contact terms. Differentiation after exterior averaging must include the covariance of the conditional means.

The Gaussian comparison. Verify the hypotheses of the normalized reference comparison on this actual chart and window. Keep the sharp-cutoff boundary estimates or the finite-window total-oscillation argument under their respective hypotheses; they are distinct routes. Establish the quantitative connected-observable errors for the actual matrices, cuts and density.

Source locations. `YM4_YMSO`: §4.12, §5.20.2, the comparison arguments in §6.1, and §6.3–6.7. The allocation, including shared passages, is in §14.13.

Bałaban references. For the reference Gaussian conditioning and covariance change, see [Ba88c, §2, (2.5)–(2.8), pp. 12–14]; for the normalized R-reference integral and Wilson cancellation, see [Ba89, §1, pp. 356–365].

FP. Short-distance asymptotics

Complete; 18 source pages.

Inputs. CP's reference-window comparison; SI's source identities and all-state transfer; GI's GD calculation and normalized free coefficient; the actual running law and coefficient bounds from RT and RC.

Required result. At the prescribed witness scale, prove the state-uniform two-point comparison

$$S_2^T(f,h) = C_G\,\zeta_K(d)^2 g_K(d)^4 \mathcal{K}(f,h)(1+r_K(f,h)), \qquad |r_K(f,h)| \le \tfrac{1}{2}, \tag{2}$$

on the separated shape class (TU9)–(TU13), uniformly in support positions and allowed states. Here K is cutoff depth, d the observation scale and S_2^T the connected physical correlation. Identify $\zeta_K(d)$ with the derivative at zero of the running coupling shift with respect to a constant scalar source, at the witness scale. Form this shift from the extracted coupling coefficients, keeping the reference data fixed during differentiation. Prove

$$g_K(d)^{-2} \sim c_* \log(1/d), \qquad \zeta_K(d) \longrightarrow 1, \qquad c_* > 0.$$

The source factor is determined by the formed coefficient maps; it is not a freely chosen normalization.

Combine the reference-window errors, physical source derivatives and free comparison. Transfer the result to all admitted Gibbs states with the same normalizers, controlling the errors relative to the g^4 signal. Use the same bare sequence and normalization throughout. The upper bound gives $d^{-8}u(d)$, with $u(d) \le C[\log(1/d)]^{-2} \to 0$, for the external rotation criterion. The positive lower bound gives nontriviality in CL.

Source locations. `YM4_YMSO`: F1–F3; §6.1–6.2, including the physical covariance transfer and identification of the running source normalization.

Bałaban references. The effective-action vacuum-polarization coefficient is analyzed in [Ba87, §5, pp. 292–297].

11. External inputs

The following general criteria are supplied externally. Their proofs are outside the construction map and its page allocation. CL must verify their hypotheses for the same limiting hierarchy and phase.

11.1. **Reflection positivity and rotations.** For translation-invariant full moment functionals, coordinate and diagonal reflection positivity, with the stated regularity and one-field insertion bounds, supplies reflection Hilbert spaces, positive translation semigroups, the coordinate spectral cone and analytic kernels. Positivity concerns full moments, not cumulants.

The rotation criterion additionally requires uniform insertion and analytic plate bounds, the exact separated shape estimate (TU10) with $u(d) \to 0$, and the angular continuation, polygon and common-cut hypotheses. Its conclusion is all-orders covariance away from coincidences. Quantitative flat-test extension transfers covariance to the stated Schwartz domain. Center a component at its own mean for the rotation argument and restore the constant mean afterwards.

Source locations. `YM4_YMSO`: G2; the Euclidean Hilbert bridge, insertion-to-plate theorem, angular two-point elimination and all-orders angular closure.

11.2. **Phases and clustering.** The phase criterion requires the reflection and moment bounds, periodic Gibbs-state realization, phase-selection hypotheses and spectral-filter estimates. It identifies the invariant part and gives decay to it with constants uniform along the limiting sequence. The selected pure component clusters. Its realization and mean must be specified; no continuum curvature-field probability law is presumed.

Source locations. `YM4_YMSO`: G3–G4; phase selection, invariant projections and clustering/filter arguments.

CL. A nontrivial scalar continuum theory

Complete; 18 source pages.

Inputs. LP, the cutoff sequence from RT, the moment/insertion bounds from SI and two-point bounds from FP, and the external criteria in §11.

Required result. At fixed spacing take the infinite-volume limit in the admitted state class. Extract a common continuum subsequence for separated correlations and construct the required Gibbs-phase realization. Prove the quantitative extension to the space $\mathcal{S}_{\neq,n}$ of Schwartz functions on $(\mathbb{R}^4)^n$ whose every derivative vanishes on each coincidence diagonal $x_i = x_j$, $i \neq j$. With $\mathcal{S}_{\neq,0} = \mathbb{C}$, the full moments G_n have $G_0 = 1$ and satisfy a bound of the form

$$|G_n(\varphi)| \leq AB^n(n!)^p|\varphi|_{kn}, \qquad |\varphi|_r = \max_{|\alpha|\leq r} \sup_x (1+|x|^2)^{r/2}|\partial^\alpha \varphi(x)|.$$

The constants have the uniformity required for phase selection.

Verify the external criteria using LP, SI and FP, and then apply them to this same hierarchy and its selected phase. The resulting phase has Euclidean invariance, permutation symmetry, reality, reflection positivity and clustering on these domains. Reflection positivity means

$$\sum_{m,n=0}^{N} G_{m+n}(\Theta\overline{\varphi_m} \otimes \varphi_n) \geq 0, \qquad \varphi_n \in \mathcal{S}_{+,n},$$

where $\mathcal{S}_{+,n} \subset \mathcal{S}_{\neq,n}$ is also flat outside the region with every time coordinate positive, and Θ reflects every time coordinate. The one-point function of the phase may be $G_1(f) = b\int f$. Keep the reference subtraction and its finite phase-dependent mean.

For the same bare sequence, source normalization and phase, pass the closed two-sided bound (2) to the selected phase. For a fixed admissible pair of nonnegative rescaled bumps of unit integral, its connected two-point function is bounded above and below by positive constants times

$$\frac{d^{-8}}{[\log(1/d)]^2}$$

for small d. Hence the scalar sector is nontrivial and has the OS properties stated in CL. A reconstruction claim requires the reconstruction theorem appropriate to these domains and growth bounds. Identification of a full Yang–Mills observable theory is the separate task F4.

Source locations. `YM4_YMSO`: Theorems 0 and 0′; D2 and H1; G1–G4; F2–F4; separated compactness, quantitative flat-test extension and component realization.

Bałaban references. The ultraviolet-stability background is [Ba88, Theorem 2 and Corollary 3, pp. 263–264] and [Ba89, Theorem 1, pp. 355–356], under their running-coupling assumptions.

13. Dependencies and outstanding estimates

The table lists the inputs beyond the shared model, geometry, domains and normalization conventions.

Box	Earlier inputs
LP	Finite Wilson model, adapted tori and admitted periodic Gibbs states
BA	Fine/coarse geometry, paths, group-valued means and stated local bounds
MB	BA; given admissible coefficient/current and constrained-Hessian data
NB	BA, MB; regular coarse data and boundary conditions
GI	BA's linear rows and MB's quadratic/operator estimates; given quadratic data
RS	BA, MB, NB, GI; one admitted input list and common running choices
EM	RS; prescribed admissible prefix and inherited remainder bounds
B	LP's finite laws, BA, MB, NB, RS, EM; actual kernels and weights
RC	BA, MB, NB, RS, EM; current-prefix bounds
LF	RS's intermediate presentation; EM's uniform bounds; B; RC admission
RT	RS, EM, B, RC, LF; initial presentation; unsourced history bounds
SI	RT, LP; finite-source formation and retained-integral estimates
CP	MB, NB, GI, RS, RT; actual histories and SI's finite-source outputs
FP	SI, CP, GI (GD), RC, RT; common physical source normalization
CL	LP, RT, SI, FP; external criteria of §11

The single-scale construction inside RT has the order

$$H_k \xrightarrow{\mathrm{RS,\,B}} T\mu_k \xrightarrow{\mathrm{RC}} \text{admitted next coupling} \xrightarrow{\mathrm{LF}} H_{k+1}.$$

The middle arrow checks coefficients of the intermediate presentation; it does not transform the measure. B supplies boundary and retained-integral estimates, including conditional estimates used inside LF's stage proof. EM supplies uniform regular-family bounds on each admitted prefix. EM and RC use the older remainder and history bounds in H_k. They do not assume completion of the next $\mathbf{R}$-operation. RT fixes constants and domains once, propagates the hypothesis, and derives the trajectory and first-passage statements.

The one-step estimates include conditional finite-source bounds. RT proves unsourced partition stability; SI verifies the sourced bounds and transfers them to Gibbs states. CP identifies the reference law and its comparison; FP derives the physical short-distance result. CL checks and applies the external criteria.

Completion in the source manuscript. The colors record completion of each box's own argument in the 1,394-page source edition identified in §14. Green means the deduction from its stated inputs is complete, including when those inputs remain unproved. Amber means part of the box's own argument remains unfinished. The existence of a proof draft or a separate manuscript does not determine the color.

LP's fixed-spacing result and BA's local averaging geometry are complete in the source. MB and GI establish their results for the given admissible coefficients and quadratic data. EM establishes its recurrence for a prescribed admissible sequence. Their green status retains those exact hypotheses.

RT's induction and partition argument, SI's source and transfer argument, FP's physical two-point deduction, and CL's continuum assembly are also complete assuming their stated inputs. In SI these include the sourced formation and retained-integral bounds, rather than only their unsourced versions. FP uses the physical source identities and the quantitative comparison. The green target records the final conditional deduction; the full construction still requires the open input boxes.

The remaining obligations are:

(i) NB requires the proper mixed-scale regular trial construction. The homogeneous curvature atlas on p. 428 does not supply it.

(ii) RS requires verification of the actual substitution domains, common cuts and summable age factor in the one-step comparison. B and LF require the actual retained-operation domains, common integrable majorants in stored coordinates, and history admission.

(iii) RC retains the actual constrained-background coefficient and Gaussian-vacuum estimates. Formal extraction and normalization formulas do not verify those estimates for the formed model.

(iv) CP requires the actual conditional-law identification and its quantitative comparison errors, including the marked normalization, conditional force and common-reference estimates; see p. 1090.

These are tasks within the corresponding boxes, beyond assuming the outputs of their predecessors. The classification uses the supplied arguments and explicit remaining tasks; it does not claim a new independent proof audit of the entire manuscript.

14. Actual source-page counts

The source is the 1,394-page edition of `YM4_YMSO` frozen on 7 November 2026: initial page A followed by printed pp. 1–1393. For an Arabic page number, add one to obtain its physical PDF page number.

Every physical page is assigned once, to its dominant mathematical task. Repeated statements, explanations and proof obligations remain counted. A mixed page belongs to one task; the other task may still cite it. The allocation below covers the entire source, without overlap or omitted pages. It measures existing source material, not the length of a future standalone manuscript or a completed proof. The former length forecasts are withdrawn.

Box	Topic	Source pages
LP	Lattice states and reflection positivity	19
BA	Block averaging and gauge geometry	43
MB	Linear lattice PDE	192
NB	Nonlinear lattice PDE	166
GI	Gaussian block integration	23
RS	One small-field step	92
EM	Iteration of regular interactions	11
B	Boundary terms and retained integrals	112
LF	The R-operation	89
RC	Counterterm calculations	40
RT	Full trajectory and partition stability	28
SI	Scalar sources and insertion bounds	51
CP	Conditional laws and Gaussian comparison	109
FP	Physical short-distance asymptotics	18
CL	Scalar continuum assembly	18
Pages assigned to the fifteen boxes		1,011
Common setup and summaries		58
External general theorems		91
Additional constructions and extensions		225
Index and references		9
Entire source		1,394

The 383 pages outside the boxes are accounted for below. Their exclusion from the diagram is a scope decision, not a claim that their arguments can be deleted. Similarly, a box's count includes its repeated exposition; no compression assumption is built into the figures.

14.1. **LP. Lattice states and reflection positivity. 19 source pages; complete.** Statement.
Printed source pages. 1, 3, 988–989, 998, 1277–1290.

The all-state proof is given in Appendices A.5–A.6, pp. 1277–1290: common-period layer removal, adapted reflection kernels, pressure comparison, separation and bounded-Borel extension. The fixed-spacing Gibbs construction and domination appear also on pp. 988–989 and 998. This status is for the admitted periodic class.

14.2. **BA. Block averaging and gauge geometry. 43 source pages; complete.** Statement.
Printed source pages. 43–56, 68–69, 76–84, 230–232, 414–417, 557, 692–694, 740, 743, 892–895, 906.

The mean, averaging, gauge and fibre-chart constructions have written proofs in §3.1–3.3, with later row and energy calculations included here. The hypotheses are the stated local small-curvature or near-identity conditions. Nonlinear background solving and later history admission are separate tasks.

14.3. **MB. Linear lattice PDE. 192 source pages; complete.** Statement.
Printed source pages. 65, 85–102, 183–200, 206–221, 224–229, 233–238, 242–246, 256–332, 369, 375–385, 403–406, 525–529, 538–540, 883–884, 886–887, 901, 908–912, 914–917, 1257–1263.

The source proves the linear operator estimates for the given admissible coefficient and current classes: coercivity, inverse bounds, regularity, local inverse families and covariance matrix functions. The later operator proofs are included, as are earlier statements of the same results. NB supplies physical Hessian identification; RS, B and LF supply their actual domain and history applications.

14.4. **NB. Nonlinear lattice PDE. 166 source pages; not closed.** Statement.
Printed source pages. 66–67, 75, 103–112, 139–182, 201–205, 222–223, 239, 247–255, 333–368, 370–374, 386–402, 407–413, 418–428, 522, 534–537, 558–559, 689–690, 864, 896–900, 907.

Extensive real and complex background proofs are written, including the homogeneous atlas in Proposition 3.193 on p. 428. The source expressly distinguishes this from a mixed-scale proper trial construction. That local construction remains required at the present box's scope.

14.5. **GI. Gaussian block integration. 23 source pages; complete.** Statement.
Printed source pages. 64, 462–464, 521, 523–524, 861–862, 866, 913, 1106–1107, 1267–1276.

The quadratic decomposition and composition, normalized complex Gaussian integral, determinant branch and moment estimates are written for given quadratic data. Appendices A.3–A.4 supply the free curvature/Wick calculation and lattice asymptotics. NB supplies the physical Hessian for applications; RS treats the interacting integral.

14.6. **RS. One small-field step. 92 source pages; not closed.** Statement.
Printed source pages. 58, 62–63, 74, 240–241, 456–461, 465–474, 477–506, 530–533, 541–548, 554–556, 562–568, 570–578, 610, 832, 841–844, 1264–1266.

Finite formation, localization and comparison arguments are written. The actual substitutions, common-cut domains and precise summable age factor remain to be verified in the one-step theorem; see pp. 832 and 841–844. These are local obligations of RS, beyond importing the linear and nonlinear background theorems.

14.7. **EM. Iteration of regular interactions. 11 source pages; complete.** Statement.
Printed source pages. 569, 715, 819, 833–834, 836–839, 928–929.

The two regular chains, weighted recurrence, squared-radius remainder sum and coefficient envelope are proved in §5.2.1. The forward induction on prescribed running data is also written on pp. 928–929. RS supplies the one-step estimate and its age factor. This result does not construct the joint trajectory.

14.8. **B. Boundary terms and retained integrals. 112 source pages; not closed.** Statement.
Printed source pages. 600–609, 611–615, 621–683, 719–724, 765–777, 786–791, 815, 827, 845–851.

The strip, read, formation, retained-kernel and complex-stage proofs are written in §4.17–4.19. The actual domains and common integrable majorant still require verification; see pp. 791–792. A fixed-chart logarithm bound alone is insufficient for that assertion.

14.9. **LF. The R-operation. 89 source pages; not closed.** Statement.
Printed source pages. 60–61, 72–73, 587–599, 616–620, 684–688, 691, 695–711, 716–718, 725–739, 741–742, 744–764, 785, 852–853.

The selected integrations, mass identities, component summation and restoration estimates have extensive conditional proofs. The actual-history application still requires the domains, stored-coordinate majorants and witness matching recorded on p. 785. The conditional component inequalities alone do not complete the stated R-operation theorem.

14.10. **RC. Counterterm calculations. 40 source pages; not closed.** Statement.
Printed source pages. 450–455, 560–561, 816–818, 825–826, 828–831, 863, 865, 867–877, 885, 888–891, 902–905, 1297.

The isotropic coefficient, determinant comparison and vacuum-accounting arguments are included. The actual constrained-background estimates and Gaussian-vacuum targets remain obligations; the target list on mixed p. 860 is counted under extensions because that page predominantly concerns a stronger comparison. This accounting decision does not remove the RC task.

14.11. **RT. Full trajectory and partition stability. 28 source pages; complete assuming inputs.** Statement.
Printed source pages. 57, 59, 70–71, 712–714, 778–780, 820–824, 835, 840, 854–858, 930–931, 986–987, 1298–1299.

The source proves conditional drift and first-passage estimates on pp. 834–835, slope and tuning statements on pp. 854–856, and normalized history and floor deductions on pp. 986–987. The formation and closure estimates left unproved on p. 835 are supplied by RS, B, LF and RC in this division. Assuming these inputs and the stated history and normalization bounds, RT's induction and partition deduction are complete.

14.12. **SI. Scalar sources and insertion bounds. 51 source pages; complete assuming inputs.** Statement.
Printed source pages. 2, 447–449, 475–476, 507–513, 792–796, 947–966, 985, 990–997, 1127–1129, 1149.

The local source energies, their indexing and source range are identified on pp. 950–951, and frozen-kernel source transport is proved on p. 952. The sourced stability, Gibbs transfer, graded moments and insertion deductions use the actual sourced formation and retained-integral estimates named in SI's inputs. Those upstream estimates remain required; their unfinished status does not leave this deduction open.

14.13. **CP. Conditional laws and Gaussian comparison. 109 source pages; not closed.** Statement.
Printed source pages. 514–520, 549–553, 967–984, 1026–1038, 1044–1093, 1099–1105, 1108–1116.

The clean-window, exterior-response, transported-covariance, source-jet and Gaussian-comparison arguments are included. The source still requires identification of the complete interacting conditional law and verification of its force, chart and marked-error bounds.

14.14. **FP. Physical short-distance asymptotics. 18 source pages; complete assuming inputs.** Statement.
Printed source pages. 1013–1025, 1039–1043.

The source-slope calculation on p. 1017 is combined with the physical source differentiation on pp. 1019–1020, which counts each pointed coupling term once. The common-normalizer and all-state transfer argument continues on pp. 1042–1043. Given CP's quantitative comparison, SI's physical source identities and the actual coefficient bounds from RC and RT, these arguments complete the two-point deduction. They use the formed model inputs, not arbitrary formal coefficient maps.

14.15. **CL. Scalar continuum assembly. 18 source pages; complete assuming inputs.** Statement.
Printed source pages. 4–5, 999–1012, 1255–1256.

The source proves conditional compactness and flat-test extension in §5.24–5.25, and assembles the phase, rotation and mean conclusions on pp. 1255–1256. Those pages retain the actual D1 and F2–F3 model inputs, supplied here by SI and FP, together with the stated external criteria. The deduction from these inputs is complete. Their remaining upstream dependencies are recorded by the open boxes rather than by CL's color.

14.16. **The 383 pages outside the boxes. Common setup and summaries: 58 pages.**

Initial page A, common norms, the main theorem and dependency catalogues, and shared analytic and geometric conventions. Repeated summaries remain counted.
Printed source pages. A, 6–42, 113–132.

External general theorems: 91 pages.

General reflection, angular continuation, reconstruction, phase decomposition, density and spectral-filter results. They remain external inputs to the diagram, as requested.
Printed source pages. 1130–1148, 1150–1157, 1172–1227, 1253–1254, 1291–1296.

Additional constructions and extensions: 225 pages.

Alternative current constructions; weighted, anisotropic and stress extensions; stronger fixed-horizon or adjacent-cutoff comparisons; Ward/displacement and full gauge-observable developments. Mixed pages may contain short passages used by a displayed box; each whole page has only one owner.
Printed source pages. 133–138, 429–446, 579–586, 781–784, 797–814, 859–860, 878–882, 918–927, 932–946, 1094–1098, 1117–1126, 1158–1171, 1228–1252, 1300–1384.

Index and references: 9 pages.

The closing index, proof-reference lists and bibliography. The opening portion of the index on mixed p. 1384 is counted with the extension that occupies most of that page.
Printed source pages. 1385–1393.

These four categories contribute $58 + 91 + 225 + 9 = 383$ pages. Together with the boxes they give $1011 + 383 = 1394$. The page assignment and the completion classification are independent: pages describing an unfinished result still count.

References

[Ba83] T. Bałaban, *Regularity and decay of lattice Green's functions*, Comm. Math. Phys. **89** (1983), 571–597. doi:10.1007/BF01214744.

[Ba84a] T. Bałaban, *Propagators and renormalization transformations for lattice gauge theories. I*, Comm. Math. Phys. **95** (1984), 17–40. doi:10.1007/BF01215753.

[Ba84b] T. Bałaban, *Propagators and renormalization transformations for lattice gauge theories. II*, Comm. Math. Phys. **96** (1984), 223–250. doi:10.1007/BF01240221.

[Ba85a] T. Bałaban, *Averaging operations for lattice gauge theories*, Comm. Math. Phys. **98** (1985), 17–51. doi:10.1007/BF01211042.

[Ba85g] T. Bałaban, *Spaces of regular gauge field configurations on a lattice and gauge fixing conditions*, Comm. Math. Phys. **99** (1985), 75–102. doi:10.1007/BF01466594.

[Ba85p] T. Bałaban, *Propagators for lattice gauge theories in a background field*, Comm. Math. Phys. **99** (1985), 389–434. doi:10.1007/BF01240355.

[Ba85v] T. Bałaban, *The variational problem and background fields in renormalization group method for lattice gauge theories*, Comm. Math. Phys. **102** (1985), 277–309. doi:10.1007/BF01229381.

[Ba87] T. Bałaban, *Renormalization group approach to lattice gauge field theories. I. Generation of effective actions in a small field approximation and a coupling constant renormalization in four dimensions*, Comm. Math. Phys. **109** (1987), 249–301. doi:10.1007/BF01215223.

[Ba88c] T. Bałaban, *Renormalization group approach to lattice gauge field theories. II. Cluster expansions*, Comm. Math. Phys. **116** (1988), 1–22. doi:10.1007/BF01239022.

[Ba88] T. Bałaban, *Convergent renormalization expansions for lattice gauge theories*, Comm. Math. Phys. **119** (1988), 243–285. doi:10.1007/BF01217741.

[Ba89a] T. Bałaban, *Large field renormalization. I. The basic step of the R operation*, Comm. Math. Phys. **122** (1989), 175–202. doi:10.1007/BF01257412.

[Ba89] T. Bałaban, *Large field renormalization. II. Localization, exponentiation, and bounds for the R operation*, Comm. Math. Phys. **122** (1989), 355–392. doi:10.1007/BF01238433.